

# CRISTIANO ULONDU MENDES

🏳️ nationality: Bissau-guinean

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## EDUCATION

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PhD in Signal and Image Processing *Palaiseau, France*

Thesis: Deep learning for Synthetic Aperture Radar (SAR) data processing and analysis *2023 -- present*

Topics: speckle filtering, polarimetric SAR despeckling, tomographic SAR reconstruction,

spatially-correlated noise, unrolled algorithms, diffusion models

*LTCI, Télécom Paris, Institut Polytechnique de Paris*

Master's Degree in Mathematics, Data, and Learning *Paris, France*

Track: Mathematics, Modeling, Learning *2021 - 2022*

*Paris Cité University*

with highest honour

Bachelor's Degree in Fundamental and Applied Mathematics *Paris, France*

*2020 - 2021*

*Paris Cité University*

with high honours

## SKILLS

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**Programming languages:** Python, MATLAB,  $\text{\LaTeX}$

**Frameworks & Libraries:** PyTorch, NumPy, SciPy, OpenCV

**Research areas:** Deep learning, inverse problems, sparse reconstruction, variational methods, SAR imaging, polarimetry, tomography

**Tools:** Git, GitLab, Linux, HPC/cluster computing

**Operating systems:** Linux, Windows

## PROJECT

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Master 1 semester project: exploration of the state of the art in inverse problem-solving methods with the aim of application in medical imaging.

Demonstrated skills:

- adaptability
- technical rigor
- autonomy and involvement

## EXPERIENCE

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- End of studies internship *01/03/22 - 31/08/22*

Laboratoire Traitement et Communication de l'Information (LTCI), Télécom Paris, France

internship subject: Exploration of variational approaches for multi-temporal despeckling of polarimetric Synthetic Aperture Radar (SAR) images.

Tasks performed :

- bibliographic study on SAR imaging

- porting the Matlab code of a single-date SAR image denoiser (MuLoG) to Python
- proposal for an extension to polarimetric SAR images of a multi-temporal denoising scheme initially developed for single-channel SAR images (RABASAR)
- proposal of solutions to account for noise spatial correlation in the despeckling of real SAR images.

Developed skills:

- organizational skills
- autonomy
- ability to analyze scientific articles, adapt, and implement the methods described therein.

- Research engineer 15/10/22 - 31/08/23  
Laboratoire Traitement et Communication de l'Information (LTCI), Télécom Paris, France

Subject: Extension of the end-of-studies internship

Tasks performed :

- writing of my first journal paper based on the internship results
- answering reviewers' comments through clarifications and additional experiments

Developed skills:

- scientific writing
- peer review response
- experimental design

## PUBLICATIONS

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### Journal Articles

- C. Ulundu Mendes, L. Denis, C. Kervazo, F. Tupin, ``Fast tomographic SAR inversion by unrolling a greedy algorithm for sparse reconstruction,'' *submitted to IEEE Transactions on Geoscience and Remote Sensing*, June 2026. [Preprint: hal-05609504]
- C. Ulundu Mendes, E. Dalsasso, L. Denis, F. Tupin, ``POLARS2POLARS: A semi-supervised despeckling algorithm for polarimetric SAR images,'' *ISPRS Journal of Photogrammetry and Remote Sensing*, 2025. DOI: 10.1016/j.isprsjprs.2025.01.008
- C. Ulundu Mendes, L. Denis, C. Deledalle, F. Tupin, ``Robustness to spatially-correlated speckle in Plug-and-Play PolSAR despeckling,'' *IEEE Transactions on Geoscience and Remote Sensing*, 2024. DOI: 10.1109/TGRS.2024.3432180

### Conferences & Workshops

- C. Ulundu Mendes, L. Denis, C. Kervazo, F. Tupin, ``Leveraging LiDAR datasets to improve SAR tomography: a diffusion model approach,'' *14th IAPR Workshop on Pattern Recognition in Remote Sensing*, Lyon, France, Aug 2026.

- C. Ulundu Mendes, L. Denis, C. Kervazo, F. Tupin, ``Reconstruction 3D en tomographie radar : apprentissage profond basé sur un Matching Pursuit déroulé,`` *GRETSI 2025*, Strasbourg, France, Aug 2025.
- C. Ulundu Mendes, L. Denis, F. Tupin, ``Schéma Plug-and-Play robuste à la corrélation spatiale du speckle pour le filtrage des données SAR polarimétriques,`` *GRETSI 2023*, Grenoble, France, Aug 2023.